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Study of temporal variation in ambient air quality during Diwali festival in India

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Abstract The variation in air quality was assessed from the ambient concentrations of various air pollutants [total suspended particle (TSP), particulate matter ≤10 μm (PM_{10}), SO_2, and NO_2] for pre-Diwali, Diwali festival, post-Diwali, and foggy day (October, November, and December), Delhi (India), from 2002 to 2007. The extensive use of fireworks was found to be related to short-term variation in air quality. During the festival, TSP is almost of the same order as compared to the concentration at an industrial site in Delhi in all the years. However, the concentrations of PM_{10}, SO_2, and NO_2 increased two to six times during the Diwali period when compared to the data reported for an industrial site. Similar trend was observed when the concentrations of pollutants were compared with values obtained for a typical foggy day each year in December. The levels of these pollutants observed during Diwali were found to be higher due to adverse meteorological conditions, i.e., decrease in 24 h average mixing height, temperature, and wind speed. The trend analysis shows that TSP, PM_{10}, NO_2, and SO_2 concentration increased just before Diwali and reached to a maximum concentration on the day of the festival. The values gradually decreased after the festival. On Diwali day, 24-h values for TSP and PM_{10} in all the years from 2002 to 2007 and for NO_2 in 2004 and 2007 were found to be higher than prescribed limits of National Ambient Air Quality Standards and exceptionally high (3.6 times) for PM_{10} in 2007. These results indicate that fireworks during the Diwali festival affected the ambient air quality adversely due to emission and accumulation of TSP, PM_{10}, SO_2, and NO_2.

Keywords Diwali · Fireworks · Air pollutants · Meteorological parameters · Mixing height · Foggy day

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Introduction

Diwali is one of the major religious festivals of India. It is generally celebrated every year during October/November over a span of few days during winter season, associated with the burning of huge amount of crackers and sparkles. However, majority of the burning is on the main day of the festival. Delhi, the capital of India, is one the most polluted cities in the world (WHO 1992). It

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has been estimated that hundreds of thousands of cases of respiratory illnesses are associated with atmospheric pollution each year in Delhi (Faiz and Sturm 2000). Large-scale fireworks lead to increase in appreciably high levels of ambient atmospheric particulates and gases like sulfur dioxide (SO_2), carbon dioxide (CO_2), carbon monoxide (CO), suspended particles (including particles below 10 μm in diameter, i.e., PM_{10}), which are associated with serious health hazards. The formation of O_3 without participation of NO_x is due to burning of sparkles (Attri et al. 2001).

Firework activities on New Year's Eve on Oahu (USA) was responsible for an increase in total suspended particulates by an average of 300% at 14 locations and by about 700% in the lung-penetrating size ranges at one location (Bach et al. 1975). Fireworks during the Diwali festival lead to a short-term variation of air quality and two to three times increase in PM_{10} and total suspended particulate concentration in Hisar city, India (Ravindra et al. 2003). On the Diwali day, 24-h average concentration was 5.7 times higher for PM_{10}, 6.6 times higher for SO_2, and 2.7 times higher for NO_x as compared to their concentration on a normal day in Lucknow city, India (Barman et al. 2008). In Oxford (England), an increase in dioxin and furan concentrations by a factor of four occurred during the period of bonfire night, suggesting that bonfires and/or fireworks could be a significant source of trace organic pollutants (Bates 1995). The use of colored sparklers by people, mostly children at ground level, put them at a high risk of inhaling the emitted pollutants.

Burning of firecrackers and sparkles increased loading effects in the human health severely especially in the infants, women, and elderly people (Kannan et al. 2004). Besides particulate matter, literature also suggests that there is strong relationship between higher concentration of SO_2 and NO_x and several health effects (Curtis et al. 2006), like cardiovascular diseases (Peters et al. 2004; Chen et al. 2005; Zannobetti and Schwartz 2002; Dockery et al. 2005), respiratory health effects such as asthma and bronchitis (Ye et al. 2001; Barnett et al. 2005), and reproductive and developmental effects such as increased risk of preterm birth (Liu et al. 2003).

In the recent years, the typical urban atmospheric environment continuously deteriorates, revealing interesting complex chemical and physical characteristics. It is also well accepted that urban air pollution episodes follow certain predetermined patterns that are associated with the local meteorological conditions and emissions of primary pollutants. Forecasting of the urban air pollution concentrations has been the topic of research of many scientific groups. In this respect, different methodologies have been developed, the majority of which are quite sophisticated and require strong infrastructure on both input parameters and computational support.

In the present study, a commercial SODAR-RASS profilometer was used to monitor the mixing height characteristics of Delhi city for a period of 6 years, i.e., 2002–2007. The objective of this paper is to correlate the profilometer estimates of mixing height with the air pollution patterns, under different meteorological conditions. In particular, the following methodology was adopted: classification of the recorded air pollution, considering primary emitted pollutants, e.g., total suspended particles (TSP), particulate matter ≤10 μm (PM_{10}), sulfur dioxide (SO_2) and nitrogen dioxide (NO_2), associated with firework events during Diwali at National Physical Laboratory (NPL), Delhi from 2002 to 2007 by collecting aerosols samples covering pre-Diwali (a day before), Diwali, post-Diwali (a day after), and foggy day. The short-term effects of these pollutants emitted from fireworks on the local environment have been discussed. To assess the daily variations in air quality, the diurnal pattern was also studied, and the ambient air quality was compared with that of a typical foggy day of December every year from 2002 to 2007. To further have a clear picture of the extent of pollution during pre-Diwali, Diwali, and post-Diwali days, a comparison with the concentration of pollutants at a typical industrial site (Shahzada Bagh) at Delhi has been made. The concentrations of pollutants for the industrial site have been taken from

Central Pollution Control Board (CPCB) website (http://www.cpcb.nic.in).

Methodology

Sampling site and description

Delhi (28°38′ N and 77°20′ E, 216 m above mean sea level), the national capital territory, occupies an area of 1,483 km^2 and has a population of nearly 14 million (Census of India 2001). Delhi's climate is mainly influenced by its inland position and the prevalence of continental air during the major part of the year. Delhi has three distinct seasons: summer, monsoon, and winter. The monsoon season is from July to September, winter months are from October to February, and summer months are from March to June. The sampling site NPL Delhi is located in central Delhi. In the vicinity, there are agricultural farms of Indian Agricultural Research Institute; area is comparatively greener than the rest of the central Delhi.

Monitoring and analysis

APM-411 (Envirotech, Delhi) and respirable dust sampler (Make, Netel Chromatographs; Model, HVS/R) were used for measuring the concentrations of TSP and PM_{10}, respectively, along with NO_2 and SO_2 in the air at a flow rate of 1.0–1.2 m^3/min for 12 h during daytime (6.0 A.M.–6.0 P.M.) and night time (6.0 P.M.–6.0 A.M.). The sampling equipment was placed on the terrace of NPL main building, i.e., 13 m from the ground level. This height aboveground level was considered as representative because usually the houses are single, double, and triple storey, and mostly, people use their roof area to burn firecrackers. Furthermore, this height can be considered as the respirable zone for people in 2/3 storey buildings. Preweighed quartz microfiber filter papers, Whatman (QMA) of 20 × 25 cm^2 sizes were used and reweighed after sampling in order to determine the mass of the particles collected. As the air with suspended particulates enters the cyclone, coarse non-respirable dust is separated from the air stream by centrifugal forces. The suspended particulate matter falls through the cyclone's conical hopper and gets collected in the cyclonic-cup. The fine dust comprising the respirable fraction of TSP passes through the cyclone and gets collected on the QMA paper. The amount of non-respirable particulate (NRP) matter and respirable particulate matter per unit volume of air passed was calculated on the basis of the difference between initial and final weights of the cyclone cup and that of the Quartz fibre filter paper and the total volume of the air sucked during sampling. Mass concentration of TSP was calculated by adding the concentration of PM_{10} and NRP, i.e., $\left[\mathrm{TSP}\left(\mu g/m^3\right) = \mathrm{PM}_{10}\left(\mu g/m^3\right) + \mathrm{NRP}\left(\mu g/m^3\right)\right]$. The experiments were performed during October, November, and December for covering the pre-Diwali (a day before), Diwali, post-Diwali (a day after), and typical foggy day (December).

A known quantity of air was passed through the impinger containing known volume of absorbing solution, and the flow rate of the impinger was set at 0.5 l/min. SO_2 was analyzed employing the West–Gaeke method (Harrison et al. 1986). Absorbing solution used for SO_2 in the impinger was potassium tetra chloromercurate (0.04 M) in distilled water. A dichlorsulphitomercurate complex is formed, which is made to react with para rosaniline and methysulphonic acid. The absorbance of the solution was measured at a wavelength of 560 nm on UV/VIS Spectrophotometer (Make: Perkin Elmer, Model: Lambda-3b). Precision, accuracy, and detection limit of the method are 4.6%, ±10.4%, and 0.75 μg SO_2 per sample, respectively. NO_2 was analyzed employing the Jacob–Hochheiser modified method (Lodge 1989). Absorbing solution used for NO_2 contained 0.4% sodium hydroxide and 0.1% sodium arsenite in distilled water. The nitrite ion thus produced was determined colorimetrically at a wavelength 540 nm by reacting the exposed absorbing reagent with phosphoric acid, sulfanilamide and *N*(1-naphthyl)ethylenediamine dihydrochloride. Precision, accuracy, and detection limit of the method are 2.6%, ±14.6%, and 1 μg NO_2 per sample, respectively.

Mixing height determination

The SODAR-RASS system was placed in the campus of NPL, Delhi. This site is considered representative of the local urban environment, from the meteorological point of view, because the low-level air flow is not interfered by topographic obstacles, thus allowing the accurate monitoring of the meteorological parameters of atmospheric boundary layer (ABL) given elsewhere by Helmis et al. (1997). The system is manufactured by REMTECH Inc. and consists of:

- A high range PA1-LR Doppler SODAR operating at 1 kHz
- A RASS system operating at a frequency of 918 MHz.

A complementary unit, the high range PA2 SODAR of REMTECH Inc., which operates at 2 kHz, is used as a backup system of the station. The system provides information on the thermal structure of the atmosphere, the wind speed and temperature profiles, the atmospheric stability and the mixing height, as well as the associated statistics up to a height of 2 km. In the present work, the mixing height was selected as a good representative of the ABL mixing properties.

Results and discussion

Burning of sulfur nitrates, magnesium, aluminium, paper, and other materials contained in crackers and fireworks produces air pollution during Diwali. The full composition and the relative concentration of the various gaseous vapors and particulate pollutants emitted are not known. However, particulate matter (especially PM_{10}), SO_2 and NO_2 are of special interest, as they are known to be potentially injurious to the respiratory passages and other health effects (Bull et al. 2001).

The meteorological parameters before Diwali and during Diwali are presented in Table 1, which shows that the meteorological conditions during monitoring periods were almost identical and can be used for the comparison of the air quality data.

Typical foggy day

Normally, fog occurs over the region after the passage of the western disturbances, as they leave sufficient moisture due to overnight rains with clear skies causing appreciable fall in night temperature. The observational study by Jenamani (2007) also confirms high frequency of fog occurrences due to very frequent western disturbances in one of the high fog month (December 1997) in contrast to few western disturbances associated with occurrences of an extreme fog month, e.g., December 2003. Delhi, north Haryana, south Punjab, and north Bihar are among the worst fog areas of India. There is evidence of a re-

Table 1 Meteorological parameters recorded during and after Diwali festival from 2002 to 2007

Year	Episode	Monitoring date	Temp[b] (°C)	RH[c] (%)	Mean wind velocity (m/s)
2002	Pre-Diwali[a]	6th October	25.2 ± 2.8	60 ± 3.9	1.6 ± 0.8
	Diwali[a]	7th October	24.1 ± 3.1	50 ± 4.1	3.6 ± 1.6
2003	Pre-Diwali[a]	24th October	26 ± 2.7	70 ± 2.2	1.2 ± 0.5
	Diwali[a]	25th October	25 ± 2.1	65 ± 1.9	4.1 ± 1.0
2004	Pre-Diwali[a]	11th November	24.1 ± 1.2	50 ± 3.4	2.1 ± 1.6
	Diwali[a]	12th November	22.2 ± 1.8	45 ± 3.5	2.8 ± 1.1
2005	Pre-Diwali[a]	31st October	23.8 ± 2.4	80 ± 1.6	3.5 ± 1.8
	Diwali[a]	1st November	23.5 ± 2.9	76 ± 1.3	4.5 ± 2.1
2006	Pre-Diwali[a]	20th October	25.1 ± 1.9	80 ± 1.1	4.1 ± 1.2
	Diwali[a]	21st October	24.5 ± 2.1	74 ± 1.7	3.6 ± 1.4
2007	Pre-Diwali[a]	8th November	23 ± 2.9	58 ± 2.1	1.7 ± 0.52
	Diwali[a]	9th November	21.1 ± 3.1	40 ± 2.9	2.1 ± 0.7

[a] Average with standard deviation
[b] Temperature
[c] Relative humidity

Table 2 Important meteorological parameters and average (24 h) ambient concentrations (μg/m^3) of pollutants in Delhi city during a typical foggy day in each year from December 2002 to 2007

Years	Monitoring date	Visibility[a] (meters)	Temperature[b] (°C)	Relative humidity[c] (%)	TSP	PM_{10}	SO_2	NO_2
2002	26th December	200	7.8	90	323[d]	122[d]	10	25
2003	23rd December	150	10.8	80	282[d]	112[d]	6	30
2004	20th December	135	8.2	85	353[d]	141[d]	12	35
2005	23rd December	293	12	80	198	86	5	18
2006	27th December	287	12	85	165	93	8	21
2007	21st December	100	6.8	93	381[d]	163[d]	16	28

[a]Eight-hour average

[b]Twelve-hour average (9 P.M.–9 A.M.) average

[c]Eight-hour average

[d]Exceeds permissible limit

lationship between pollution, land use patterns, and fog frequency. In urban setups like Delhi, the high level of pollutants in the air provides abundant surfaces for the water vapor to condense to form fog (Chung and Ramanathan 2006). The average concentration of TSP, PM_{10}, SO_2, and NO_2 on the typical foggy days in each December month from 2002 to 2007 at NPL are presented in Table 2 along with the important meteorological parameters and compared with the concentration of pollutants measured on the Diwali day. These measurements have been done each year keeping in view the high concentration of pollutants on a typical foggy day. The comparison of results for foggy day with those during Diwali would help to assess the extent of pollution due to the festival.

Health effects of pollutants

The higher concentration of PM_{10} in inhaled air has more possibility of these particles to reach deep in lungs. Entry of particles into lungs is dangerous because they can carry a complex mixture of toxic pollutants from fireworks. Michie et al. (2000) have also studied the effects of PM_{10} particles on pediatric respiratory diseases following community fireworks. There are reports on the decrements in peak expiratory flow rates in children, in relation with PM_{10} concentration (Hoek et al. 1998).

Reviews and analysis of epidemiological literature for acute adverse effects to particulates (TSP and specifically PM_{10} or smaller particles) have been studied by Pope et al. (1995), Schwartz (1993), and Dockery et al. (1993). COEHA (1996) estimated these effects and found that increase of PM_{10} by 10 μg/m^3 increases the mortality rate percent and have been related to premature death, aggravated asthma, increased hospital admission, and increased respiratory problems (Table 3). Individuals who were elderly or had pre-existing

Table 3 Combined effect estimates of daily mean particulate pollution (COEHA 1996)

Health impacts	Change (%) in health indicator per each 10 μg/m^3 increase in PM_{10}
Increase in daily mortality	
Total deaths	1
Respiratory deaths	3.4
Cardiovascular deaths	1.4
Increase in hospital usage (all respiratory diagnoses)	
Admissions	1.4
Emergency department visits	0.9
Exacerbation of asthma	
Asthmatic attacks	3
Bronchodilator use	12.2
Emergency department visit	3.4
Hospital admissions	1.9
Increase in respiratory symptom report	
Lower respiratory	3
Upper respiratory	0.7
Cough	2.5
Decrease in lung function	
Forced expiratory	0.15
Peak expiratory flow	0.08

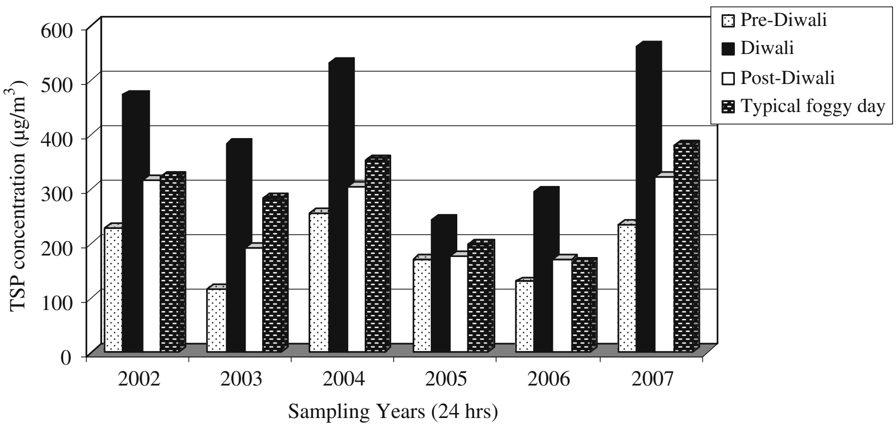


Fig. 1 Gives the variation in the concentration of total suspended particulate matter (TSP) during Diwali and typical foggy day at the monitoring site from 2002 to 2007

lung or heart disease appeared to be more susceptible than others to adverse effects of PM_{10}.

High levels of SO_2 are particularly dangerous in the presence of particulate matter because it slowly adsorbs on fine atmospheric particles and can be transported very deep into lungs and therefore staying there for a long time. Due to their very long residence time and acidic character, they can cause serious damage to the lung tissue (edema). Dockery et al. (1993) have shown that among the particles of diverse composition, sulfates have the worst health impact, which also stay in air for long time. Gong et al. (1995) studied the short-term health response to SO_2 exposure on asthmatic patients. They found that a 10-min SO_2 exposure at concentration >1.32 mg/m^3 and ventilation 30 dm^3/min can cause short-term asthma manifestations more intense than those usually experienced from everyday stress without SO_2 exposure.

NO_2 is a deep lung irritant, which has been shown to generate biochemical alterations and histological demonstrable lung damage in laboratory animals as a result of both acute and chronic exposure Hoek et al. (1998). Ponka and Virtanen (1994) have found a significant association with emergency room visit for exacerbation of chronic bronchitis or emphysema, with high levels of air pollutants specifically SO_2 and NO_2. Inhaled NO_2 can penetrate to small lung airways, and hence, there is much greater susceptibility with NO_2 to broncho-constrictive response in individuals with asthma (Schindler et al. 1998; WHO 2000).

Air pollutants-TSP, PM_{10}, SO_2, and NO_2

The variation of TSP, PM_{10}, SO_2, and NO_2 during 3-day measurement periods covering pre-Diwali (1 day before), Diwali (the festival day) and post-

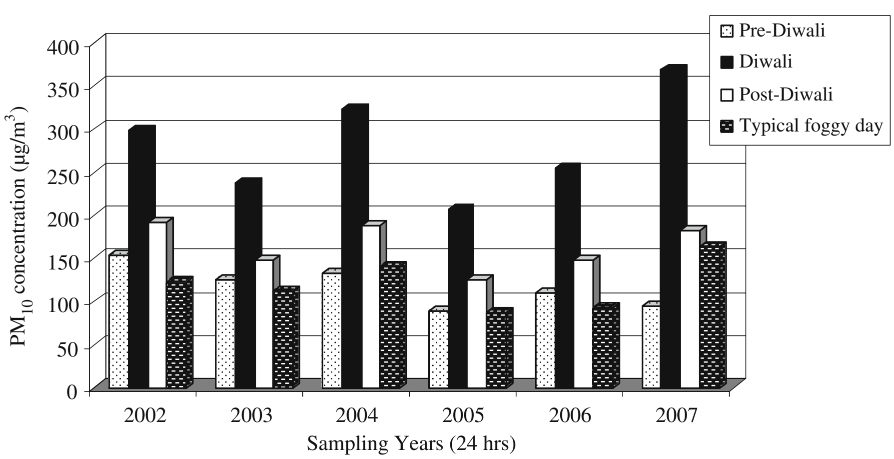


Fig. 2 The variation in the concentration of respirable particulate matter, i.e., PM_{10} during Diwali and typical foggy day at the monitoring site from 2002 to 2007

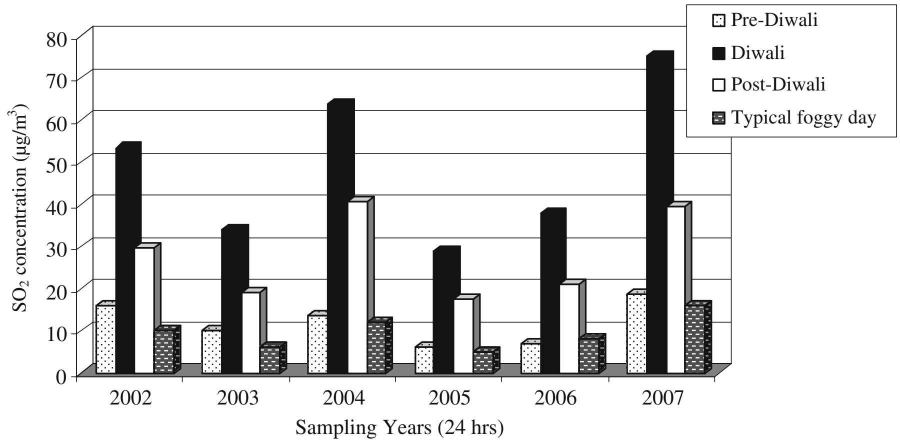


Fig. 3 Gives the variation in the concentration of sulphur dioxide (SO_2) in the ambient air during Diwali and typical foggy day at the monitoring site from 2002 to 2007

Diwali (a day after) are shown in Figs. 1, 2, 3, and 4, respectively. A sharp rise in TSP, PM_{10}, SO_2, and NO_2 was observed on Diwali, i.e., the day of the festival, because tremendous amount of crackers and sparkles are burnt on this very day. Before and after the festival, the level of these species is much lower.

The 24-h concentrations of four parameters TSP, PM_{10}, SO_2, and NO_2 for monitoring period, i.e., 2002–2007, were found to be several times higher than the normal day and the prescribed National Ambient Air Quality Standards (NAAQS). A large increase of TSP, PM_{10}, SO_2, and NO_2 than the normal day values has been reported by Central Pollution Control Board (CPCB 2000) Delhi, with maximum 30 times increase of SO_2 during Diwali in Delhi, India, and exposure to this deadly gas is responsible for the incidence of severe respiratory diseases (http://cseindia.org/campaign/apc/dark_trends.htm).

Total suspended particle

TSP concentration exceeded the maximum prescribed limits (Table 4) in all years from 2002 to 2007 during Diwali for residential areas. Before Diwali, the daily average concentration ranged from 117.5 to 235 μg/m^3 in the monitoring years. On Diwali, the maximum concentration of 563 μg/m^3 was noted in 2007, which is almost 2.5 times than the maximum prescribed limit (Table 4) during the period 2002–2006. After Diwali, the TSP concentration slightly decreased, but it is also above the prescribed limit in 2002, 2004, and 2007. Babu and Moorthy (2001) have also observed a large increase in the black carbon concentration by a factor of over 3 above the unperturbed background level at a remote coastal location, associated with the Diwali festival in 2000. Overall, the TSP on pre-Diwali day is almost around the prescribed limits (Table 4), but on

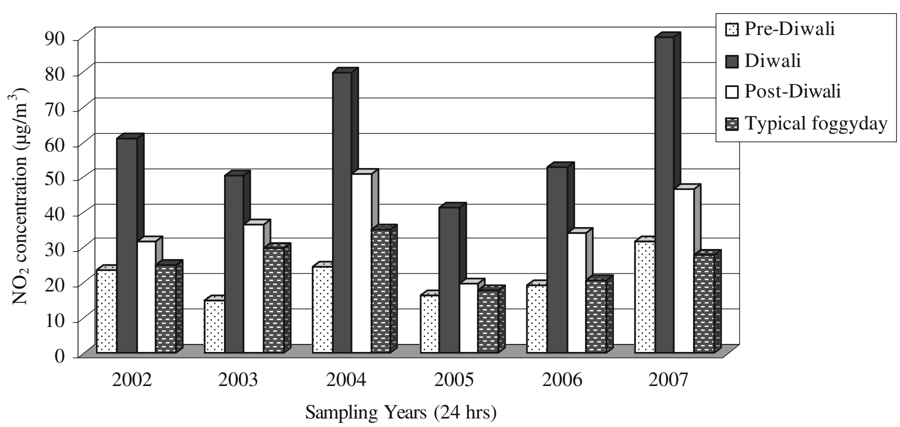


Fig. 4 The concentration of nitrogen dioxide (NO_2) in the ambient air during Diwali and typical foggy day at the monitoring site from 2002 to 2007

Table 4 National ambient air quality standards (CPCB 2000)

Pollutant	Concentration in ambient air ($\mu g/m^{3a}$)		
	Sensitive area[b]	Industrial area	Residential area
TSP	100	500	200
PM_{10}	75	150	100
SO_2	30	120	80
NO_2	30	120	80

[a]Time weighed average (24 h)

[b]Hospitals, schools, colleges, institutes, etc

the day of Diwali, it has increased as high as three times during the years under observation. On comparing the TSP concentrations during Diwali with the TSP concentration reported at an industrial site (Shahzada Bagh) in Delhi (Table 5) at Shahzada Bagh, it can be seen that the TSP concentrations are of the same order and much above the prescribed limits. The TSP concentrations on the day of Diwali are even greater than the concentrations measured for a typical foggy day at NPL in each year, as can be seen in Fig. 1.

Particulate matter ≤10 μm

Concentration of PM_{10} every year exceeded the maximum prescribed allowable limit before and after Diwali except in year 2005 and 2007. During Diwali, a two to four times increase in PM_{10} concentration was observed as compared to its concentration on pre-Diwali. A slight fall in PM_{10} concentration was observed a day after the festival (Fig. 2). It has been observed that, during diwali period, the concentration of PM_{10} is almost two times higher than the concentration reported at an industrial site (Shahzada Bagh). The PM_{10} level on Diwali (Fig. 2) is much higher (more than two times) than the foggy day, which is monitored each year during December, although the meteorological conditions were very similar. Clark (1997) has also reported a dramatically short-term (hourly average) increase in the particle level, which originated from bonfires and fireworks across England and Wales. Furthermore, the PM_{10} levels were strongly influenced by local conditions, i.e., how close the sites were to large bonfires, and weather changes.

Table 5 Concentrations of pollutants ($\mu g/m^3$) at Shahzada Bagh (Industrial area) of Delhi from 2002 to 2007

Year	TSP[a]	PM_{10}^a	SO_2^a	NO_2^a
2002	468 ± 188	187 ± 53	9.7 ± 1.3	33.9 ± 6.5
2003	375 ± 130	140 ± 25	6.9 ± 2.1	30.3 ± 10.4
2004	340 ± 140	179 ± 54	12 ± 3.2	36.8 ± 8.6
2005	245 ± 152	133 ± 32	5.6 ± 1.4	32.1 ± 4.6
2006	278 ± 138	123 ± 36	4.3 ± 0.3	30.1 ± 6.7
2007	516 ± 165	197 ± 51	15 ± 2.8	39.9 ± 5.6

Data taken from CPCB website: http://www.cpcb.nic.in/

[a]Annual average concentration

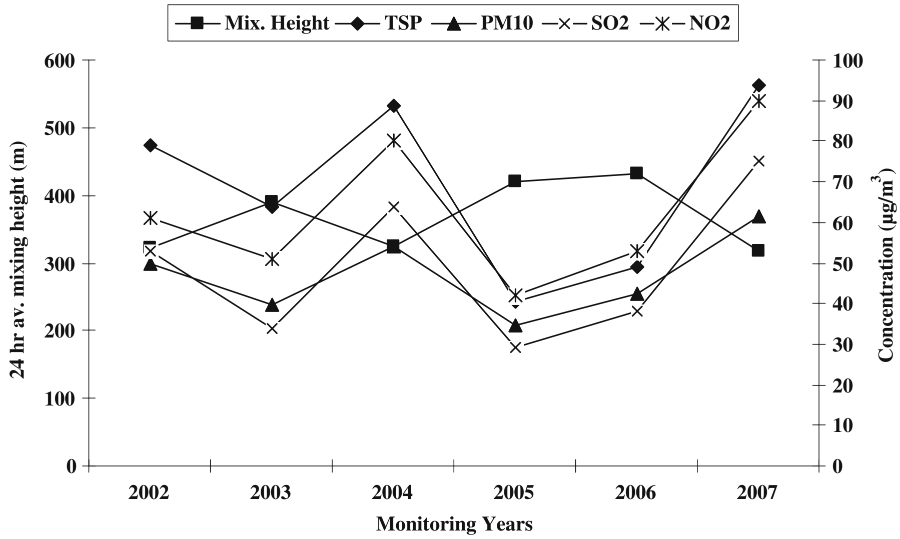
Sulfur dioxide

In general, firecrackers contain 75% potassium nitrate, 15% carbon (C), and 10% sulfur (S). Potassium nitrate is a strong oxidizing agent when burnt with C and S; it releases gases such as CO_2 and SO_2. The daily average concentration of SO_2 before Diwali varied from 6 to 18 $\mu g/m^3$. During Diwali, the SO_2 concentration ranged from 29 to 75 $\mu g/m^3$, i.e., almost three to six times higher than the daily average concentration. On comparing with industrial emissions (Table 5), it was observed that SO_2 concentration on the day of Diwali was almost five to six times higher. A day after Diwali, the SO_2 concentration decreased significantly, as compared to the levels on Diwali (Fig. 3). The SO_2 concentration on the day of diwali is three to six times higher than typical foggy day (Table 2).

Nitrogen dioxide

In 2004 and 2007, NO_2 concentration exceeded the maximum permissible limit (Table 4). On Diwali, two to four times increase in NO_2 concentration was observed compared to that at an industrial site. Even on post-Diwali, the average NO_2 concentration was found to be higher than the emissions at an industrial site (Table 5). Two to four times increase in NO_2 concentration was also observed when compared to the NO_2 concentration for a typical foggy day (Fig. 4). On comparing the cumulative data from 2002 to 2007, it is observed that TSP, PM_{10}, SO_2, and NO_2 val-

Fig. 5 Gives changes in concentration of various air pollutants with change in mixing height from 2002 to 2007



ues are very high during 2002, 2003, 2004, and 2007. This increase may possibly be attributed to adverse meteorological conditions [i.e., decrease in 24 hourly average temperature, average wind speed (Table 1), and average mixing height (Fig. 5)] on Diwali days, which has caused accumulation of pollutants at lower level, thereby increasing the concentration of all air pollutants substantially. Adverse meteorological conditions are generally observed during beginning of winter season (November), and the conditions were quiet similar on the day of Diwali during 2002, 2003, 2004, and 2007. TSP, PM_{10}, SO_2, and NO_2 recorded lower trend in 2005 and 2006 as compared to 2002, 2003, 2004, and 2007 possibly because of bursting of lesser crackers due to anti-cracker campaigns launched by the regulatory/concerned authorities like Ministry of Environment and Forest, CPCB, Delhi government, NGOs, schools, newspapers, and electronic media. Significant reduction in air pollution level in these 2 years (i.e., 2005 and 2006) is also possibly attributed to favorable meteorological conditions [increase in mixing height (Fig. 5) and high wind speed], hence better dispersion of pollutants and rains on previous day of Diwali in 2006 (CPCB 2006).

The increase of TSP and PM_{10} were much more than SO_2 and NO_2, indicating that changes in the meteorology alone cannot explain the observed changes. Burning of fireworks is the strong source of suspended particles. Large amount of soot in them catalyze atmospheric reactions (Wang et al. 2006). It has been observed that average mixing height (Fig. 5) has decreased by 18% and 30% in 2004 and 2007 as compared to 2003, 2005, and 2006, where average mixing height has increased by 19%, 26%, and 2% respectively; therefore, TSP, PM_{10}, SO_2, and NO_2 levels at NPL increased during 2004 and 2007 as compared to 2003, 2005, and 2006. Lower mixing height results in less volume of troposphere available for mixing, resulting in build-up of air pollutants near the surface and hence resulting in higher concentrations. Thus, one of the probable reasons for the increase in TSP, PM_{10}, SO_2, and NO_2 levels during 2004 and 2007 as compared to 2003, 2005, and 2006 is the decrease in mixing height.

Diurnal pattern

The diurnal pattern for the concentrations of TSP and PM_{10} do not show significant changes during day and night (Table 6). On Diwali night, the concentrations of TSP, PM_{10}, SO_2, and NO_2 were higher than their respective daytime levels in all the monitoring years. Furthermore, the daytime concentration of TSP, PM_{10}, SO_2, and NO_2 on Diwali day was found to be significantly higher

Table 6 Diurnal pattern of determined air pollutants during Diwali festival[a]

Pollutant		2002			2003			2004		
		Night[b]	Day[c]	24 h[d]	Night[b]	Day[c]	24 h[d]	Night[b]	Day[c]	24 h[d]
TSP ($\mu g/m^3$)	Pre-Diwali	241	214.6	227.8 ± 18.7[e]	123	112	117.5 ± 7.8[e]	286	226	256 ± 42.4[e]
	Diwali	493.2	452.7	472.9 ± 28.6[e]	355	412	383.5 ± 32.3[e]	562	502	532 ± 42.4[e]
	Post-Diwali	324.7	308.6	316.7 ± 11.4[e]	209	174	191.5 ± 24.7[e]	324	287	305.5 ± 26.2[e]
PM_{10} ($\mu g/m^3$)	Pre-Diwali	190	116	153 ± 52.3[e]	139	112	125.5 ± 19.1[e]	148	116	132 ± 22.6[e]
	Diwali	332.3	264.9	298.6 ± 47.7[e]	276	198	237 ± 55.2[e]	345	312	323.5 ± 23.3[e]
	Post-Diwali	205	177.6	191.3 ± 19.4[e]	165	132	148.5 ± 23.3[e]	175	201	188 ± 18.4[e]
SO_2 ($\mu g/m^3$)	Pre-Diwali	19	13	16 ± 4.2	13	7	10 ± 4.2	18	9	13.5 ± 6.4
	Diwali	70.6	36	53.3 ± 24.5	46	21.5	33.8 ± 17	72	55	63.5 ± 12
	Post-Diwali	36	23	29.5 ± 9.2	13	25	19 ± 8.5	42	39	40.5 ± 2.1
NO_2 ($\mu g/m^3$)	Pre-Diwali	31	16	23.5 ± 10.6	18	12	15 ± 4.2	28	21	24.5 ± 4.9
	Diwali	87	35.5	61.3 ± 36.4	56	45	50.5 ± 7.8	85	75	80 ± 7.1[e]
	Post-Diwali	25.7	38	31.9 ± 8.7	31	42	36.5 ± 7.8	65	47	51 ± 12.7

		2005			2006			2007		
		Night[b]	Day[c]	24 h[d]	Night[b]	Day[c]	24 h[d]	Night[b]	Day[c]	24 h[d]
TSP ($\mu g/m^3$)	Pre-Diwali	185	156	170.5 ± 20.5[e]	141	119[e]	130 ± 15.6	256	214	235 ± 29.7[e]
	Diwali	278	207	242.5 ± 50.2[e]	312	278	295 ± 24[e]	588	538	563 ± 35.4[e]
	Post-Diwali	197	159	178 ± 26.9[e]	189	153	171 ± 25.5	352	292	282 ± 42.4[e]
PM_{10} ($\mu g/m^3$)	Pre-Diwali	98	78	88 ± 14.1	138	82	110 ± 39.6[e]	106	83	94.5 ± 16.3
	Diwali	246	167	206.5 ± 55.9[e]	278	232	255 ± 32.5[e]	388	348	368 ± 28.3[e]
	Post-Diwali	135	115	125 ± 14.1	158	136	147 ± 15.6[e]	198	165	181.5 ± 23.3[e]
SO_2 ($\mu g/m^3$)	Pre-Diwali	7	5	6 ± 1.4	9	5	7 ± 2.8	23	14	18.5 ± 6.4
	Diwali	35	23	29 ± 8.5	45	31	38 ± 9.9	85	65	75 ± 14.1
	Post-Diwali	21	11.5	16.3 ± 6.7	25	17	21 ± 5.7	48	31	39.5 ± 12
NO_2 ($\mu g/m^3$)	Pre-Diwali	19	27	23.5 ± 5.7	26	13	19.5 ± 9.2	38	26	32 ± 8.5
	Diwali	45	38	41.5 ± 4.9	60	46	53 ± 9.9	98	82	90 ± 11.3[e]
	Post-Diwali	22	18	20 ± 2.8	45	23	34 ± 15.6	55	38	46.5 ± 12

[a]Pre-Diwali = 24 h before Diwali; Diwali = the night and the following day; post-Diwali = 24 h after Diwali

[b]Night: 6 P.M. to 6 A.M.

[c]Day: 6 A.M. to 6 P.M.

[d]24 h average with standard deviation

[e]Exceeds permissible limits

Table 7 Correlation among TSP, PM_{10}, SO_2, and NO_2 for monitoring period 2002 to 2007

Parameters	TSP	PM_{10}	SO_2	NO_2	RH%	WS	Temp.
TSP	1	0.99**	0.97**	0.96**	0.99**	0.92**	0.81*
PM_{10}		1	0.93**	0.94**	0.98**	0.88**	0.75*
SO_2			1	0.96**	0.97**	0.87**	0.85**
NO_2				1	0.95**	0.91**	0.89**
RH%					1	0.94**	0.87**
WS						1	0.92**
Temp.							1

$^*p < 0.05$, $^{**}p < 0.01$

(almost two to six times) than the previous daytime (pre-Diwali) concentration (Table 6). In general, there were no fireworks during daytime, and besides, due to a public holiday, the source of vehicular pollution might have been less than the pre-Diwali day. Even then, the increase in concentration indicated a longer residence time of these pollutants in the ambient air accumulated on pre-Diwali night due to fireworks. Diurnal pattern of TSP, PM_{10}, SO_2, and NO_2 concentration showed a slight increase in night (6 P.M. to 6 A.M.) as compared to daytime (Table 6), which seems to be associated with increased firework events during the night (a special characteristic of Diwali night).

Relationship between TSP, PM_{10}, SO_2, NO_2, and meteorological parameters

The relationship between TSP, PM_{10}, SO_2, NO_2, and meteorological parameters (temperature, wind speed, and humidity) in 2002–2007 Diwali periods was analyzed by linear regression analysis. The correlation coefficient (R) between 24-h average TSP, PM_{10}, SO_2, NO_2, and meteorological parameters are shown in Table 7. As can be seen from the table, the correlation of SO_2 and NO_2 with meteorological parameters is very much similar to the correlation between TSP, PM_{10}, and meteorological parameters. During the study period, the variation of 24-h simultaneous TSP and PM_{10} concentration data was very well correlated ($r = 0.99$), and this correlation was statistically significant at the 95% confidence ($p < 0.01$; Table 7). The variation of TSP and PM_{10} concentrations compared to their SO_2 and NO_2 concentrations was highly correlated ($r = 0.97$ and $r = 0.96$ and $r = 0.93$ and $r = 0.94$, respectively), and this correlation was statistically significant at the 95% confidence ($p < 0.01$; Table 7). TSP, PM_{10}, SO_2, and NO_2, compared with relative humidity, temperature, and wind speed, were very well correlated, and this correlation was statistically significant at the 95% confidence ($p < 0.01$ and $p < 0.05$; Table 7).

It is obvious that the concentration of pollutants decrease effectively with increasing temperature, wind speed, and relative humidity. This situation shows that, when wind speed is high, pollutants dilute by dispersion.

Conclusions

This study shows that the burning of crackers and sparkles on the occasion of Diwali is a strong source of TSP, PM_{10}, SO_2, and NO_2 in ambient air and are emitted in very high quantity, as high as two to six times, as compared to non-Diwali festival days. However, the high concentrations decrease sharply within the next 48 h, indicating their accumulation for few hours only in the ambient air. Increase in TSP, PM_{10}, SO_2, and NO_2 values in general may be attributed to adverse meteorological conditions, i.e., decrease in 24 hourly average temperature, 24 hourly average wind speed, and 24 hourly average mixing height on Diwali days in 2002, 2003, 2004, and 2007 from the previous day, which has caused accumulation of pollutants at lower level, thereby increasing the concentration of all air pollutants substantially. All the 24-h average concentrations of TSP, PM_{10}, SO_2, and NO_2 were found to be higher than the NAAQS. The higher level of pollutants, especially the manifold increase (four times) of PM_{10} is of great concern with regard to the health effects. Strong correlation was observed between TSP, PM_{10}, SO_2, and NO_2 and meteorological parameters (temperature, relative humidity, and wind speed).

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